EXPERIMENT 1: TITANIC DATASET PREPROCESSING AND VISUALIZATION

AIM:

To perform data preprocessing and exploratory data analysis on the Titanic dataset by inspecting data, handling missing values and duplicates, encoding categorical data, scaling numerical data, and visualizing feature relationships.

ALGORITHM:

1. Import the required Python libraries: Pandas, Seaborn, Matplotlib, and scikit-learn preprocessing tools.
2. Load the Titanic CSV dataset into a Pandas DataFrame and inspect the first rows, data types, and missing values.
3. Fill missing Cabin values with “Unknown” and remove duplicate records.
4. Encode the Sex column using LabelEncoder.
5. Standardize the Fare column using StandardScaler.
6. Fill missing Age values using forward fill and backward fill.
7. Plot selected features (Pclass, Sex, Age, SibSp) using a pair plot.
8. Calculate and display the correlation heatmap for Pclass, Age, SibSp, Parch, and Fare.

Step 1: Load the Titanic Dataset and Convert it into a DataFrame

```
[ ]
#EXP_NO_1
#SANJAY_S_241801246
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.preprocessing import LabelEncoder
from sklearn.preprocessing import StandardScaler

from google.colab import drive
drive.mount('/content/drive')

file_path = "/content/drive/MyDrive/PODS/Titanic-Dataset.csv"
df = pd.read_csv(file_path)
```

Step 2: Explore and Understand the Dataset

Display the first few rows

```
]
#EXP_NO_1
#SANJAY_S_241801246
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

Get information about column data types and missing values

```
#EXP_NO_1
#SANJAY_S_241801246
df.info()
print("\nMissing Values:")
print(df.isnull().sum())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   PassengerId  891 non-null    int64  
 1   Survived     891 non-null    int64  
 2   Pclass       891 non-null    int64  
 3   Name         891 non-null    object 
 4   Sex          891 non-null    object 
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64  
 7   Parch        891 non-null    int64  
 8   Ticket       891 non-null    object 
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object 
 11  Embarked     889 non-null    object 
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB

Missing Values:
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

Step 3: Do Forward Fill and Backward Fill on 'Age'

```
[ ]
#EXP_NO_1
#SANJAY_S_241801246
df["Age"] = df["Age"].ffill()
df["Age"] = df["Age"].bfill()
print(df["Age"].isnull().sum())
```

```
0
```

Step 4: Fill Missing Values in 'Cabin' with "unknown" (Limit = 5)

```
[ ]
#EXP_NO_1
#SANJAY_S_241801246
df["Cabin"] = df["Cabin"].fillna("unknown", limit=5)
print(df["Cabin"].isnull().sum())
```

```
682
```

Step 5: Remove Duplicate Records (if any)

```
[ ]
#EXP_NO_1
#SANJAY_S_241801246
print("Duplicate Records:", df.duplicated().sum())
df = df.drop_duplicates()
print("Dataset Shape:", df.shape)
```

```
Duplicate Records: 0
Dataset Shape: (891, 12)
```

Step 6: Encode Categorical Column 'Sex' using LabelEncoder

```
#EXP_NO_1
#SANJAY_S_241801246
encoder = LabelEncoder()
df["Sex"] = encoder.fit_transform(df["Sex"])
df[["Sex"]].head()
```

	Sex
0	1
1	0
2	0
3	0
4	1

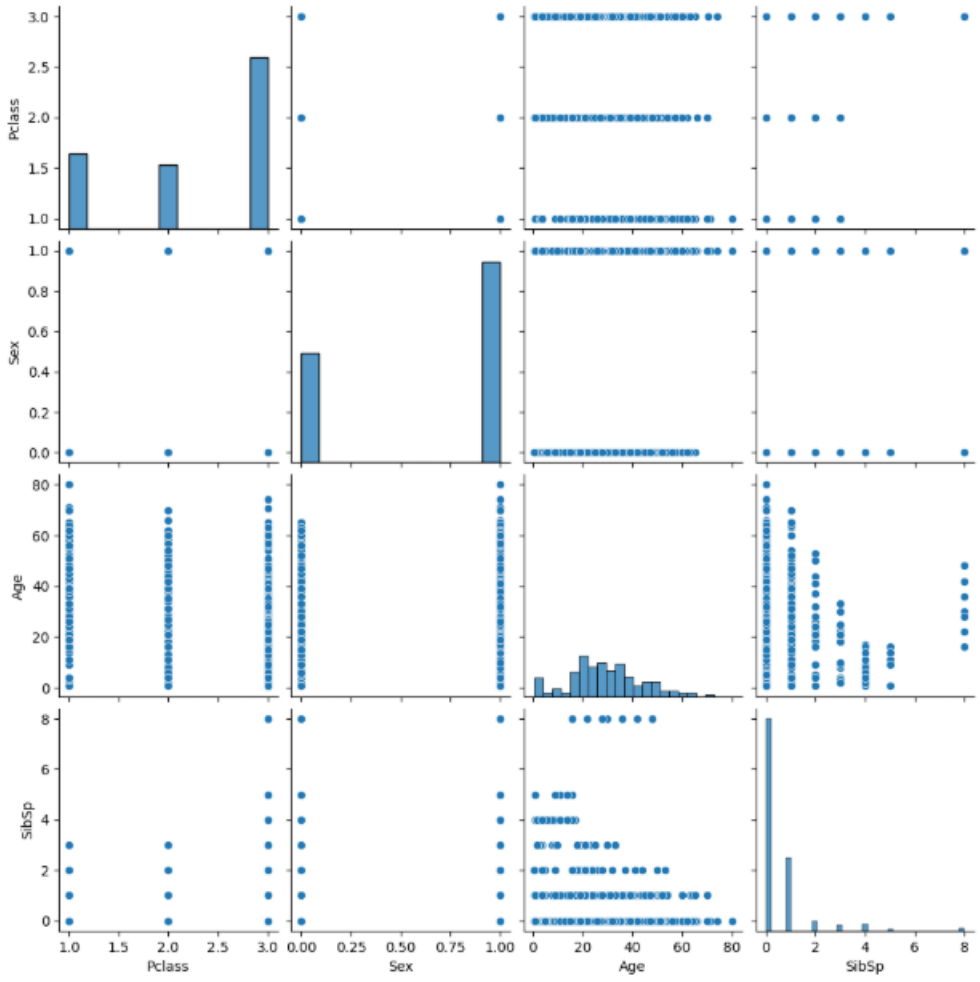
Step 7: Scale Numerical Feature 'Fare' using StandardScaler

```
#EXP_NO_1
#SANJAY_S_241801246
scaler = StandardScaler()
df["Fare"] = scaler.fit_transform(df[["Fare"]])
df[["Fare"]].head()
```

	Fare
0	-0.502445
1	0.786845
2	-0.488854
3	0.420730
4	-0.486337

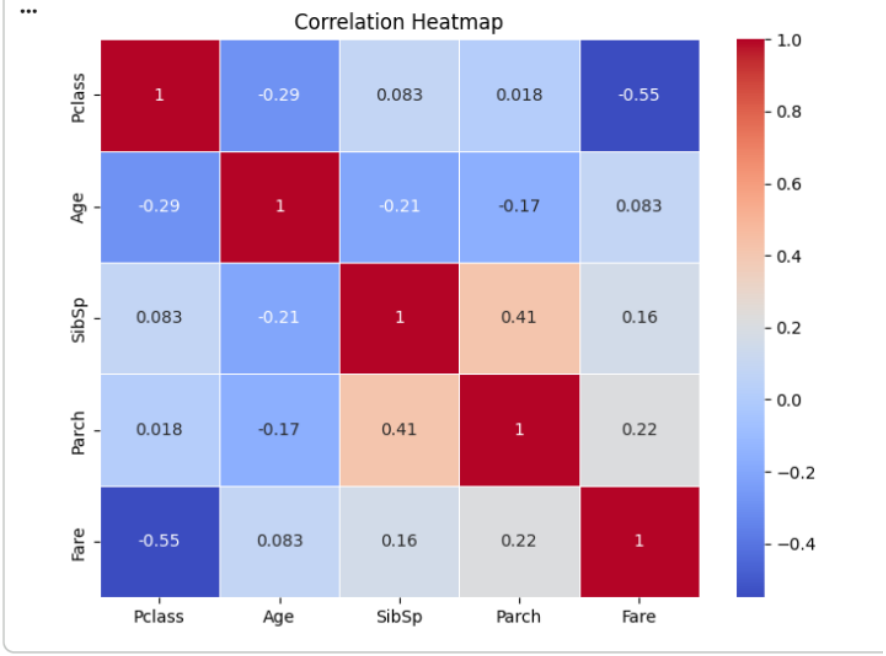
Step 8: Pair Plot of Selected Features ('Pclass', 'Sex', 'Age', 'SibSp')

```
#EXP_NO_1
#SANJAY_S_241801246
sns.pairplot(df[['Pclass', 'Sex', 'Age', 'SibSp']])
plt.show()
```



Step 9: Display the Correlation Heatmap ('Pclass', 'Age', 'SibSp', 'Parch', 'Fare')

```
#EXP_NO_1
#SANJAY_S_241801246
plt.figure(figsize=(8,6))
corr = df[['Pclass', 'Age', 'SibSp', 'Parch', 'Fare']].corr()
sns.heatmap(corr, annot=True, cmap="coolwarm", linewidths=0.5)
plt.title("Correlation Heatmap")
plt.show()
```



RESULT:

The Titanic dataset was loaded into a DataFrame and explored. Missing values were handled, duplicate records were removed, the Sex column was label-encoded, the Fare column was standardized, and visualizations (pair plot and correlation heatmap) were generated to examine feature distributions and relationships.